

Supplementary Material for “Conflict-Complete Retrieval: Checkable Contradictions under a Context Budget”

Anonymous submission

Scope

This supplement expands proofs and records the reviewer-facing reproduction contract. It does not introduce an additional headline claim; the main paper remains self-contained. Core results concern deterministic synthetic fixtures; the semi-natural leg preserves public gold linkage topology and natural attributes but injects the single-valued conflict predicate. Neither supplies a natural-prevalence estimate or independently adjudicated semantic conflict labels.

Assumption Ledger

The guarantees use the following assumptions.

1. The premise store is finite after applying the confidence threshold. Assertions have exact RDF-term identity and retain their source identifiers.
2. Each keyed class has one supplied key predicate. Key equality is exact; no entity linker, string normalization, or probabilistic match is inferred.
3. Only schema-declared single-valued predicates can generate the conflicts studied here. Time intervals must overlap and scopes must agree unless an annotation is explicitly unspecified.
4. Query relevance is determined by supplied entity seeds and requested predicates. Conflict completeness requires one complete minimal witness for every relevant conflict, not every alternative derivation. Context cost is monotone under set inclusion.
5. The maintained index is evaluated on the current insertion-oriented snapshot. Deletion, persistence, concurrent mutation, and crash recovery are outside the implementation contract.
6. Exact joint selection enumerates simple identity paths and searches their witness combinations. Path and search-node caps can force safe SEARCH-LIMIT; only an exhaustive run claims optimality or infeasibility.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

Expanded Correctness Arguments

Detector Soundness and Completeness

Let direct co-keying connect two typed records that share the exact key value, and let its reflexive, symmetric, transitive closure be $\sim_K$. Propagation copies only schema-constrained values within one $\sim_K$ class. Therefore a derived unequal-value pair has endpoints in one semantic entity class and satisfies the predicate, confidence, time, and scope guards. This proves soundness.

For completeness, take any semantic conflict. Its endpoint records are in one finite $\sim_K$ class, so a finite path of direct co-key steps connects them. Repeated propagation along the path carries both constrained values to a common representative, where the unequal-value rule fires. The finite source leaf set of that derivation contains a minimal subset: repeatedly delete a leaf whenever entailment remains. Termination is immediate because the leaf set is finite. The remaining subset is a witness by definition.

Joint Budget Feasibility

For a finite relevant conflict set Γ , choose one witness $W_\gamma \in \mathcal{W}_\gamma$ for each conflict and take their union. Finiteness of the premise store makes the collection of possible unions finite, so a minimum-cost union U^* exists. If $\mu(U^*) \leq B$, returning U^* is conflict-complete and respects the budget. Conversely, any conflict-complete context R contains some witness W_γ for every γ . Their union is a subset of R ; monotonicity gives $b^* \leq \mu(R) \leq B$. This proves the if-and-only-if boundary.

Let $\widehat{W}_\gamma$ be the shortest witness independently selected by CWI and $\widehat{U} = \bigcup_\gamma \widehat{W}_\gamma$. Under triple count, each $\widehat{W}_\gamma$ is no larger than the witness for γ contained in U^* , and every such witness is no larger than U^* . Hence

$$|\widehat{U}| \leq \sum_{\gamma \in \Gamma} |\widehat{W}_\gamma| \leq |\Gamma| |U^*| = |\Gamma| b^*.$$

The implementation deduplicates $\widehat{U}$ before testing the budget. It returns the entire union if it fits; otherwise it returns SELECTION-OVERFLOW and an empty context. The qualifier “selection” is essential because a different combination of non-shortest witnesses could have a smaller union.

For completeness, the joint-selection decision problem is in NP when witness families are explicit: a certificate

chooses one witness per conflict, and its union and size are polynomially checkable. For NP-hardness, reduce Set Cover with universe E and candidate sets $A_1, \dots, A_m$. Create conflict γ_e and private token d_e for every $e \in E$, shared token a_j for every candidate set, and witness family $\mathcal{W}_{\gamma_e} = \{\{d_e, a_j\} : e \in A_j\}$. Every joint union contains the $|E|$ private tokens. Its remaining tokens name candidate sets, and choosing one available witness per element is exactly choosing sets that cover E . Thus $b^* \leq |E| + k$ if and only if a cover of size at most k exists.

Exact Joint Selection

Represent a direct co-key assertion pair as a record–bucket–record edge. Any minimal identity proof connecting two conflicting endpoint records can contain no cycle: deleting a cycle edge preserves endpoint connectivity and contradicts minimality. It therefore corresponds to a simple path in the bipartite record–bucket graph. Conversely, the type and key premises on any simple path, together with the endpoint values and required annotations, form an inclusion-minimal witness: removing any path premise disconnects that certificate, and removing an endpoint value removes the violation.

The exact selector depth-first enumerates these paths, deduplicates equal source-triple sets, and performs branch-and-bound over one witness choice per conflict. The default caps are 10,000 paths per conflict and 1,000,000 search nodes. The independently shortest union initializes the upper bound. When enumeration and selection finish, every tuple in the finite Cartesian product $\prod_{\gamma \in \Gamma} \mathcal{W}_\gamma$ has either been visited or pruned because its current union is already no smaller than a complete incumbent. The incumbent therefore has size b^* . A fitting incumbent is safe even if a cap is reached; a non-fitting incumbent justifies INFEASIBLE only after exhaustive search. Otherwise the implementation returns SEARCH-LIMIT and an empty context.

Fixed-Hop Non-Type IRI Incompleteness

Fix any finite hop limit k . Construct typed records x and y whose key objects are the same literal, whose constrained values disagree, and for which no non-`rdf:type` triple has an IRI-object path between x and y . A query seeded at x has a relevant direct-key conflict because exact literal-key equality places the records in one $\sim_K$ class. Its logical witness contains the two type, two key, and two value assertions.

The non-type IRI graph omits literals as vertices and does not traverse class-membership edges. Consequently x and y are in different connected components, not merely more than k hops apart. Any policy restricted to the k -hop component of x omits the type, key, and value premises incident to y and cannot contain the witness. Since the construction works for every finite k , no fixed finite radius gives the claimed guarantee for this policy class. Literal-aware, key-aware, and type-traversing policies are outside the proposition. This final exclusion is substantive: if `rdf:type` edges are traversed, records of the common class C are connected through C in two hops.

Family	Instances	Conflicts	Mechanism
D20	14	8	same/direct key
XR-small	16	12	direct key + distractors
XR-scale	64	60	scaled direct key
Chain-2	41	10	two co-key links
Chain-3	25	6	three co-key links
H3-NK1	16	12	hierarchy/non-key
H3-NK3	16	12	more non-key predicates
TauSig	60	12	time/scope guards
Total	252	132	

Table 1: The deterministic core synthetic fixture matrix.

CWI Contract and Witness Minimality

After all assertions in a snapshot have been processed, a record belongs to a key bucket exactly when its type and key premises are present. Union–find therefore represents precisely the connected components of the record–bucket graph, which are the $\sim_K$ classes. Within each class, the index compares only schema-constrained values and applies the same time/scope guard as the formal semantics. Hence its materialized conflict set is sound and complete for the finite snapshot.

For a reported cross-record conflict, breadth-first search selects a shortest record–bucket path between the two endpoint records. The certificate contains both endpoint values and, for every path edge, the type and key premises that justify the edge. Removing either endpoint value removes the unequal-value premise. Removing any path premise breaks that co-key edge; because the path is shortest and the certificate contains no alternative path, the endpoints are no longer connected inside the certificate. Thus no proper subset entails the conflict. A chain of m direct co-key links has $m + 1$ type assertions, $2m$ key assertions, and two endpoint values, for $3m + 3$ logical assertions. A same-record conflict needs only the two unequal values.

Insertion order does not change the final snapshot relation. A record enters a bucket only once its required type and key are both known; a later premise triggers the same membership update that an earlier premise would have triggered. Later constrained values and annotations re-evaluate their touched class. The tests exercise permutations in which types, keys, values, and annotations arrive before and after one another.

For a multi-seed query, the budget wrapper canonicalizes conflicts reached from more than one seed, filters the requested predicate footprint, and deduplicates source assertions by stable triple identifier. Focused tests verify shared-premise deduplication, exact-budget success, one-below-budget overflow with no partial context, predicate filtering, alternative-path enumeration, exact union minimization, proven infeasibility, and capped-search abstention.

Evaluation Matrix

For every instance, the shipped manifest records the query, expected label, fixture identifier, and stable source data. Exact and maintained systems are scored by a checker over the

Family	Exact witness	Key-aware	BM25	Dense	Hybrid
D20	129	260	489	505	480
XR-small	479	623	6,383	26,639	12,527
XR-scale	479	623	623	130,319	24,815
Chain-2	718	7,395	34,519	34,519	34,519
Chain-3	957	7,157	26,447	26,447	26,447
H3-NK1	479	679	7,279	29,999	14,319
H3-NK3	479	801	17,311	37,319	34,812
TauSig	735	879	16,152	48,311	31,637

Table 2: Mean UTF-8 bytes of query-returned assertions at the same smallest fully conflict-complete settings as the main retrieval table. The shared oracle schema is excluded. The artifact also reports maxima and the selected parameter for every cell.

g	Shared sum	Shared b^*	Disjoint b^*
1	6	6	6
2	12	8	12
4	24	12	24
8	48	20	48
16	96	36	96
32	192	68	192

Table 3: Exact triple costs in the unique-path joint-budget fixtures.

assertions they return. A positive counts only when the returned context contains a complete source-grounded witness. Negative controls include different entities, shared non-key values, disjoint time intervals, and unequal scopes.

The retrieval table in the main paper selects the smallest tested configuration that reaches complete recall within a family. The tested record-level depths are 1–4; relevance methods use $K \in \{1, 2, 4, \dots, 512\}$. Reciprocal-rank fusion uses 60. Means and maxima are over conflict queries. The scale companion uses one unrecorded warm-up and 11 measured repetitions and reports medians and interquartile ranges. These are descriptive system measurements, not statistical-significance claims.

Multi-Conflict Budget Matrix

The shared family places g conflicting predicates on the same two directly co-keyed records. Its four type/key premises are shared and each conflict adds two values, giving $b^* = 4 + 2g$. The disjoint family uses g separate record pairs and multiple query seeds, giving $b^* = 6g$. For every row, the shipped runner checks both $B = b^*$ and $B = b^* - 1$.

The alternative-path fixture contains four conflicts. Each has a private two-link shortest path and four alternatives, one of which traverses a shared three-link spine. Independent shortest selection uses 33 triples. Exhaustive selection over 20 candidates visits 226 search nodes, proves a 27-triple optimum, returns complete evidence at $B = 27$, and proves infeasibility at $B = 26$. Capping each family at one enumerated path instead yields SEARCH-LIMIT, not a false infeasibility claim.

Persons	q	CWI	Cached	Fresh
300	1	1.18	0.98	0.95
300	100	1.62	0.99	102.94
3,000	1	8.89	19.21	17.00
3,000	100	9.25	19.21	1,765.49
10,000	1	40.97	84.35	79.02
10,000	100	41.53	84.36	7,909.10

Table 4: Median total milliseconds over 11 repetitions after one warm-up. Cached exact is a fair immutable-snapshot comparator; fresh exact is the no-retained-state endpoint.

Semi-Natural External Matrix

The external runner reads `tableA.csv`, `tableB.csv`, and `matches.csv` from the DBLP–Google Scholar entity-resolution benchmark, verifies their recorded SHA-256 digests, and constructs components from supplied gold match edges. Selection is lexicographic and component-disjoint: rare depth-four components are reserved first, followed by depths three, two, and one, with a target of 20 pairs per depth. The available data yield 20 pairs at depths 1–3 and eight at depth 4.

For each pair, all component records retain nonempty title, author, venue, and year fields as non-constrained distractor assertions. Each gold edge is encoded as a unique exact key value shared by its endpoints. The runner adds one controlled single-valued predicate: unequal values give a positive and equal values give its paired negative. Across 136 instances, CWI returns all 68 positive witnesses, rejects all 68 negatives, and every witness is rechecked against source assertions. Components range from 2 to 37 records; contexts from 12 to 224 triples; natural distractors from 6 to 113; and positive witnesses from 6 to 15 triples. The labels validate the retrieval mechanism on natural topology, not the prevalence of semantic conflicts.

Maintenance Workload Matrix

For each of 300, 3,000, and 10,000 persons, the generator and schema parser run before timing. One unrecorded warm-up precedes 11 measured repetitions. In a repetition, CWI ingests the in-memory snapshot once and answers the first $q \in \{1, 10, 100\}$ deterministic conflict queries. Cached exact batch-builds its triple index, exact key closure, conflict cells, and subject lookup once per repetition, then reuses them. Fresh exact reconstructs the same exact structures for every query. All arms receive the same in-memory triples, so none receives a file-I/O advantage.

Every query has exactly one expected conflict and no run is missing or spurious. Full per-run timings and IQRs are shipped. CWI retains 1,386–46,006 structural/conflict items (0.820–0.821 per source triple). Cached exact retains 1,869–62,009 source-index, detection, and lookup references; these representation-specific counts are not heap-byte estimates. Fresh exact retains no derived state between queries.

Large-Class Stress Matrix

This adversarial leg measures the disclosed worst regime rather than store-size scaling: the maintained pair set is

n	Triples	Pairs	Ingest ms, median [IQR]
8	24	28	0.095 [0.059, 0.107]
16	48	120	0.336 [0.289, 0.555]
32	96	496	0.505 [0.382, 0.587]
64	192	2,016	3.014 [2.751, 3.370]
128	384	8,128	19.266 [18.584, 20.006]
256	768	32,640	253.058 [245.788, 264.722]

Table 5: Median [IQR] over 11 runs after one warm-up when all records share one key and have distinct constrained values.

exactly $n(n - 1)/2$. Timings are descriptive and machine-dependent; pair and triple counts are deterministic.

Artifact Contract

The anonymous code/data archive contains:

- TypeScript source and focused tests for the maintained index, exact comparators, heuristic and exact joint-budget queries, adaptive and on-demand witnesses, workload comparison, retrieval policies, instance scoring, and scale ledger;
- all eight synthetic fixture families and their instance manifests;
- frozen paper-facing outputs for exact, maintained, on-demand, sparse, dense, curve, joint-budget, semi-natural, workload, class-stress, and scale analyses;
- the pinned Python environment and runner for the dense baseline; and
- the external topology and semi-natural scripts with frozen reports.

The archive deliberately omits repository history, internal planning documents, hosted-reader runs excluded from the paper, service credentials, caches, installed dependencies, compiled output, and submission-production files. Dense re-execution requires obtaining the cited third-party encoder weights; frozen dense outputs are included so the reported table remains auditable without that download. Timing will vary by machine and should be regenerated only as a systems check, not expected to match byte for byte. For double-blind review, the project-owned RDF namespace is replaced consistently in source, fixtures, and frozen textual outputs by a reserved `example.org` namespace. This name-only substitution changes no graph structure, label, measurement, or scientific value.

Additional Limitations

Independent shortest-witness selection need not minimize the joint union; exact selection resolves this only when exponential path enumeration and branch-and-bound finish below their caps. Materialized unequal-value pairs can be quadratic within a large equivalence class; the dedicated stress leg exposes that regime. The semi-natural conflict predicate is injected, and unequal natural publication-year strings remain unlabeled because versions, linkage errors, or metadata errors could explain them. The workload uses a single process and an insertion-batch/read-many schedule. Cached exact

is an immutable-snapshot comparator; the study does not cover concurrent or interleaved updates, deletion, or heap-residency measurement.